William Bartel

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EDUCATION

The University of Notre Dame Notre Dame, IN
Ph.D., Mechanical Engineering Expected May 2028
Robotics and Controls Group - ROAM Lab
Advisor: Patrick Wensing

Villanova University Villanova, PA
B.S., Mechanical Engineering, May 2023
Concentration: Dynamics,
magna cum laude

RESEARCH EXPERIENCE

University of Notre Dame Notre Dame, IN
Generative AI Applications to Legged Robots, ROAM Lab June 2025 – present

- Designing physics-guided flow matching framework, leveraging classic control theory, robotics fundamentals, and modern generative AI strategies toward the completion of humanoid loco-manipulation tasks
- Constructing motion capture data collection pipeline for recording human expert demonstrations to be used in imitation learning

Quadruped Loco-Manipulation Research, ROAM Lab September 2023 – June 2025

- Implemented control of quadruped robot with arm to reject disturbances in simulation using a Cascaded Fidelity Model Predictive Control framework in C++
- Coded trajectory optimization of single rigid body model of quadruped robot with an arm using iLQR in MATLAB

Villanova University Villanova, PA
VESL Research, Department of Mechanical Engineering May 2022 – July 2022

- Aided in designing and constructing a functioning robot created to drag mine detectors under vegetation to allow safer removal of landmines in Cambodia through the Villanova Engineering Service Learning (VESL) program,
- Designed electronics housing, soldered new boards, updated system schematics, wired the robot, and updated code
- Robot was then used by PhD students for further research

TEACHING EXPERIENCE

University of Notre Dame Notre Dame, IN

Lead Instructor, First-Year Engineering 2026-2027

- Developed curriculum, delivered lectures, and assessed students as instructor of record for two consecutive semesters

Teaching Assistant, Department of Aerospace and Mechanical Engineering

Mechatronics Lab Spring 2026

- Instructing and guiding 16 students through their lab assignments each week
- Building community, taking attendance, and grading

Introduction to Robotics Fall 2025

- Gave a guest lecture, gave a robot demonstration with my labmate and fellow TA, and helped prepare class projects and tools
- Tutored students through office hours twice a week and graded work

Introduction to Design Thinking 2023-2024

- Graded assignments and provided students with feedback on their work

PUBLICATIONS AND PRESENTATIONS

Conference Paper 2026

W. Bartel et al., "EulerFlow: Inverse-Dynamics-Guided Flow Matching for Humanoid Motion Generation"

Accepted — IEEE-RAS International Conference on Humanoid Robots (Humanoids 2026)

Presentations

Physics-Guided Flow Matching for Humanoid Loco-Manipulation February 2026

Notre Dame Physical AI Poster Session, Notre Dame, IN

The Robotic Future of Indiana Manufacturing February 2026

7th Annual Indiana Science Communication Day, Indianapolis, IN

HONORS & AWARDS

First-Year Engineering Teaching Apprenticeship Program Fellowship 2025-2026

University of Notre Dame AME Graduate Scholarship 2023

Villanova National Merit Scholarship 2020

TECHNICAL SKILLS

Programming: Python, Pytorch, C++, Java - competent

Software: MATLAB, SOLIDWORKS - proficient

Hardware: Robot design, construction, soldering - competent

LEADERSHIP & OUTREACH

Community and University Outreach 2024-present

- Over the summers present my research and conduct robot demonstrations for high school students considering engineering
- Gave a tour of the ROAM Lab to group of local students involved with robotics teams
- Presented my research and demonstrated the quadruped robot from the ROAM Lab to families of the Notre Dame Engineering Advisory Council
- Volunteered as a judge for the high school division of the Northern Indiana Regional Science and Engineering Fair

Assistant Rector, O'Neill Family Hall men's dorm 2025-2026

- Fostered community and served residents through men's group programming, supervising student-led activities, and looking out for their well-being
- Two nights a week was in charge of the 260 young men in the dorm and was the contact for any problems - building maintenance, a student's health, interpersonal conflicts, etc.

Robotics Head Coach, Robinson Community Learning Center 2024

- Led two assistant coaches in teaching robotics, coding, and team skills to 8 students between the ages of 8 and 10
- The Robinson Community Learning Center provides programs and resources for students who largely qualify for free or reduced lunch and promotes STEM opportunities to students from underrepresented groups